

Differential genome-wide associated variants and enriched pathways of ECG parameters among people with versus without HIV

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Objectives: People with HIV (PWH) are more likely to develop ECG abnormalities. Substantial evidence exists for genetic contribution to ECG parameters among general population. However, whether and how would host genome associate with ECG parameters among PWH is unclear. Our research aims to analyze and compare genetic variants, mapped genes, and enriched pathways of ECG parameters among PWH and HIV-negative controls.

Design: A cross-sectional study.

Method: We performed a large original genome-wide association study (GWAS) of ECG parameters among PWH ($n=1730$) and HIV-negative controls ($n=3746$). Genome-wide interaction analyses were also conducted.

Results: A total of 18 novel variants were detected among PWH, six for PR interval including rs76345397 at *ATL2*, 11 for QRS duration including rs10483994 at *KCNK10* and rs2478830 at *JCAD*, and one for QTc interval (rs9815364). Among HIV-negative controls, we identified variants located at previously reported ECG-related genes (*SCN5A*, *CNOT1*). Genetic variants had a significant interaction with HIV infection ($P < 5 \times 10^{-8}$), implying that HIV infection and host genome might jointly influence ECG parameters. Mapped genes for PR interval and QRS duration among PWH were enriched in the biological process of viral genome replication and host response to virus, respectively, whereas enriched pathways for PR interval among HIV-negative controls were in the cellular component of voltage-gated sodium channel complex.

Conclusion: The present GWAS indicated a distinctive impact of host genome on quantitative ECG parameters among PWH. Different from HIV-negative controls, host genome might influence the cardiac electrical activity by interfering with HIV viral infection, production, and latency among PWH.

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Introduction

Antiretroviral therapy (ART) has superbly improved the prognosis of people with HIV (PWH). As a consequence, risks of aging-related diseases such as cardiovascular diseases (CVDs) and cardiometabolic comorbidities are increasingly rising and become the major cause of death among PWH [1]. Previous studies have indicated that PWH are more likely to develop cardiac complications, including atherosclerosis, arrhythmia, myocardial infarction, sudden cardiac death, and heart failure when compared with HIV-negative controls [2,3].

ECG is valuable and reliable in cardiovascular assessment and an essential tool for arrhythmia and various cardiac disorders diagnosis [4]. The measurements routinely obtained with ECG include PR interval, QRS duration, and QT interval, which are indicative of the function of the conduction system and provide important prognostic information [5]. PR interval corresponds to the sum of atrial and atrioventricular nodal conduction, the disturbance of which is related to atrial fibrillation risk [6]. The QRS complex reflects ventricular depolarization, and increased QRS duration can lead to life-threatening bradyarrhythmias and tachyarrhythmias [7]. The QT interval reflects myocardial repolarization and is a risk marker for ventricular arrhythmia and sudden cardiac death [8]. Studies have shown that the prevalence of ECG abnormalities was higher among PWH compared with general population, but mechanisms behind them were still unclear [1].

Substantial evidence exists for the host genetic contribution to the cardiac conduction system with a reported heritability for QT interval between 35–40% [9,10] and 34% for the PR interval [9,11]. Also, several genome-wide association studies (GWAS) indicated that variants at *NOSA1P*, *SCN5A*, *CNOT1*, *KCNQ1* genes were associated with PR interval, QRS duration, or QT interval among the general population [12–15]. However, genetic association study on ECG parameters among PWH was lacking. HIV virus could cause inflammation, integrate into host genome, and produce HIV latency reservoir, which might cause different genetic association with ECG parameters. Whether and how would the common genetic variants associate with ECG parameters differentially between PWH and general population remains to be clarified.

Therefore, in the present study, we conducted a large GWAS of ECG parameters, including PR interval, QRS duration, and QT interval among Chinese PWH and HIV-negative controls in an attempt to compare the differences of genetic associations with ECGs between these two groups. As host and pathogen variants jointly determine disease outcomes, it is necessary to seek approaches that allow the simultaneous assessment of their respective and combined effects [16]. Thus, the potential effects of HIV infection and integration of proviral DNA into host

genome were assessed by genome-wide interaction analyses. Uncovering the genetic architecture of ECG parameters among PWH can help dissect pathogenic mechanism and development of arrhythmia and other cardiac disorders.

Materials and methods

Study design and participants

Comparative HIV and Aging Research in Taizhou (CHART) study is an ongoing prospective cohort study of HIV and aging-related comorbidities among PWH and HIV-negative controls in Eastern China [17]. The present cross-sectional study was based on the baseline survey of CHART conducted in 2017–2020.

For the present analysis, we selected the individuals with at least one parameter on ECG, including PR interval, QRS duration, and QT interval. Written informed consent was obtained from all study participants. The study was approved by the Institutional Review Board of Fudan University School of Public Health, Shanghai, China.

Data collection

A standardized structured questionnaire was administered face-to-face by trained health staff to collect information on age, sex, tobacco and alcohol use, and history of noncommunicable diseases (NCDs). Details and definitions can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

ECG measurements

A standard 12-lead ECG was performed on all individuals at supine position after 5 min rest using standardized procedures. ECG parameters were automatically measured using the integrated 12SL-Code. A trained ECG technician conducted and inspected the process. Heart rate correction was performed by the Bazett formula to obtain the QT-corrected (QTc) interval. Details can be seen Supplemental materials, <http://links.lww.com/QAD/C924>.

Genome-wide association analyses

Genomic DNA, extracted from whole peripheral blood samples using a commercial DNA extraction kit (QIAamp DNA Blood Kits; Qiagen, Hilden, Germany), and quantified using PicoGreen reagent (PicoGreen dsDNA Quantitation Reagent; Invitrogen, Carlsbad, California, USA), was genotyped on the Infinium Chinese Genotyping Array-24 v1.0 BeadChip (727 349 SNPs) [18]. SNPs and individuals used in the GWAS were selected according to the following criteria: individual call rate more than 95%, SNP call rate more than 95%, minor allele frequencies at least 0.05, and Hardy–Weinberg equilibrium at least 1×10^{-6} . Principal component analysis was done for the remaining 386 420 SNPs and the first five principal components were extracted and employed in association analyses.

GWA analyses were performed under additive genetic effects assumption. The association between ECG parameters and genetic variants was measured by linear mixed model (LMM). LMM-based methods are usually preferred largely because they can account for population stratification [19] and relatedness without the need to remove related individuals [20]. LMM was conducted through fastGWA model implemented in the GCTA software package [21,22]. For dichotomous ECG parameters, generalized linear mixed model (GLMM) was used [23].

The genome-wide significance threshold was considered at P value less than 5×10^{-8} , and P value less than 1×10^{-6} indicated a suggestive significance threshold. Details can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

Statistical analysis

Histograms of ECG parameters were reviewed for normality. Comparisons of baseline characteristics, stratified by HIV serostatus, were performed using Student's t -test or Chi-square test in appropriate. Generalized linear regressions were conducted to examine the association of baseline characteristics and ECG parameters. A P value less than 0.05 served as statistical significance. Data were analyzed with SAS 9.4 software (SAS Institute, Cary, North Carolina, USA).

In order to test the interaction between HIV infection and host genome on ECG parameters, we also conducted a genome-wide interaction analysis by including a two-way interaction parameter based on the equation: $g(Y) = \beta_0 + \beta_1 \times \text{SNP} + \beta_2 \times \text{HIV in fection} + \beta_3 \times (\text{SNP} \times \text{HIV in fection}) + \beta_4 \times X_c$. The coefficient β_3 was the interaction effect of HIV infection and genetic variants on ECG parameters. Details can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

Functional annotation, gene mapping, and gene set analysis

We applied Functional Mapping and Annotation of genetic associations (FUMA) [24] with predefined values to detect individual significant SNPs and corresponding genomic risk loci. Functional consequences for these SNPs were obtained by performing ANNOVAR [25]. SNPs that were genome-wide significant or were in linkage disequilibrium (LD) ($r^2 > 0.6$) with one of the independent significant SNPs were mapped to genes in FUMA using one of three strategies: positional mapping, eQTL mapping, and chromatin interaction mapping.

Genes indicated by mapping of the SNPs were further investigated using the GENE2FUNC procedure in FUMA. The adjusted P value (FDR) for gene set enrichment analysis was calculated by the Benjamini-Hochberg method and the threshold was 0.05. The minimum number of input genes overlapping with a

tested gene set was two. Bubble plots of enriched pathways (ER) were illustrated by R package 'ggplot2'. Details can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

Results

Demographic and ECG characteristics

In total, 1730 PWH and 3746 HIV-negative individuals were included. The two groups had comparable age distribution, but a higher proportion of men was observed in PWH (Table 1).

PWH had a shorter mean PR interval (151.58 vs. 158.07 ms, $P < 0.001$) and QRS duration (95.07 vs. 96.58 ms, $P = 0.015$) than HIV-negative counterparts but had a much longer mean QTc interval (419.54 vs. 412.09 ms, $P < 0.001$) and resting heart rate (78.40 vs. 71.89 bpm, $P < 0.001$) (Table 1). The histograms of PR interval, QRS duration, and QTc interval subordinated an approximately normal distribution (Supplemental figure 1–3, <http://links.lww.com/QAD/C924>). Other details can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

Correlates of ECG parameters

HIV infection was negatively associated with PR interval ($\beta = -4.83$, $P < 0.001$), QRS duration ($\beta = -1.14$, $P = 0.032$) but positively associated with QTc interval ($\beta = 7.78$, $P < 0.001$) in multivariable generalized linear model, respectively. Hypertension was the only parameter positively associated with all of the three ECG manifestations after adjustment (PR interval: $\beta = 3.15$, $P = 0.001$; QRS duration: $\beta = 1.24$, $P = 0.039$; QTc interval: $\beta = 3.25$, $P = 0.001$) (Supplemental table 1, <http://links.lww.com/QAD/C924>). Details can be seen in Supplemental materials, <http://links.lww.com/QAD/C924>.

Genetic variants associated with ECG parameters

A total of 386 420 SNPs and 5476 participants were subject to final association analyses of ECG parameters (Supplemental figure 4, <http://links.lww.com/QAD/C924>, Table 2). Manhattan and qq plots of ECG parameters among PWH are shown in Fig. 1 with λ ranging around 0.997 to 1.013, indicating the well controlled genomic inflation and no population stratification. Manhattan and qq plots among HIV-negative controls can be seen in Supplemental figure 5–10, <http://links.lww.com/QAD/C924>.

PR interval

Among PWH, six variants at six genes reached genome-wide significance level to PR interval (rs1144157, $\beta = 16.55$, $P = 5.05 \times 10^{-10}$, MAP3K7; rs115038157, $\beta = 16.17$, $P = 4.11 \times 10^{-9}$, MAFA; rs17043654, $\beta = 15.76$, $P = 6.12 \times 10^{-9}$, RPL23AP39; rs73643283,

Table 1. Sociodemographic characteristics and electrocardiogram abnormalities among study participants.

Characteristics	PWH (n = 1730)	HIV-negative controls (n = 3746)	P ^a
Age (years)			0.021
18~29	362 (20.92)	653 (17.43)	
30~44	555 (32.08)	1254 (33.48)	
45~59	502 (29.02)	1121 (29.93)	
60~89	311 (17.98)	718 (19.17)	
Male	1347 (77.86)	2780 (74.21)	0.004
BMI (kg/m ²)			<0.001
<18.5	159 (9.19)	206 (5.50)	
18.5–24.0	1133 (65.49)	1696 (45.27)	
>24	438 (25.32)	1844 (49.23)	
Smoking status			<0.001
Never	1044 (60.35)	2021 (53.95)	
Previous	215 (12.43)	391 (10.44)	
Current	471 (27.23)	1334 (35.61)	
Regular alcohol use			<0.001
Yes	85 (4.91)	561 (14.98)	
No	1645 (95.09)	3185 (85.02)	
PR interval (ms, mean ± SD) (n = 5447)	151.58 ± 33.31	158.07 ± 25.43	<0.001
QRS duration (ms, mean ± SD) (n = 5457)	95.07 ± 24.39	96.58 ± 12.37	0.015
QTc interval (ms, mean ± SD) (n = 5344)	419.54 ± 33.90	412.09 ± 26.12	<0.001
PR interval abnormality	75 (4.37)	206 (5.52)	0.075
QRS interval abnormality	44 (2.54)	129 (3.44)	0.077
QTc interval abnormality	533 (31.32)	697 (19.14)	<0.001
Resting heart rate (bpm, mean ± SD) (n = 5438)	78.40 ± 13.76	71.89 ± 10.89	<0.001
Sinus tachycardia	79 (4.57)	48 (1.28)	<0.001
Sinus bradycardias	82 (4.74)	304 (8.12)	<0.001
ST/T wave abnormalities	218 (12.60)	355 (9.48)	<0.001
Any ECG abnormality	541 (31.27)	1247 (33.29)	0.139
cIMT category (mm) (n = 5408)			<0.001
<0.78	983 (59.11)	2580 (68.89)	
0.78–1.00	437 (26.28)	773 (20.64)	
>1.00	243 (14.61)	392 (10.47)	
Dyslipidemia (n = 5455)			<0.001
Yes	1021 (59.53)	1933 (51.68)	
No	694 (40.47)	1807 (48.32)	
Hypertension			<0.001
Yes	381 (22.02)	1219 (32.54)	
No	1349 (77.98)	2527 (67.46)	
Diabetes			<0.001
Yes	119 (6.88)	400 (10.68)	
No	1611 (93.12)	3346 (89.32)	
Metabolic syndrome			<0.001
Yes	507 (29.31)	1337 (35.69)	
No	1223 (70.69)	2409 (64.31)	
CD4 ⁺ cell count (cells/μl) (n = 2563)			
≤200	601 (37.54)		
201–350	591 (36.91)		
>350	409 (25.55)		
Years since HIV diagnosis>3 years	556 (32.14)		
Years of ART>2 years (n = 1588)	595 (37.47)		

ART, antiretroviral therapy; cIMT, carotid intima media thickness; PWH, people with HIV.

^aCompared between PWH and HIV-negative counterparts, assessed by chi-square test for categorical variables and Student's *t*-test or Wilcoxon sign rank test for continuous variables.

$\beta = 14.02$, $P = 8.89 \times 10^{-9}$, *LINGO2*; rs76345397, $\beta = 14.15$, $P = 4.06 \times 10^{-8}$, *ATL2*; rs9941936, $\beta = 14.29$, $P = 4.60 \times 10^{-8}$, *ENTHD1*). Another three variants at *ATL2* gene (rs60941472, rs17022549, rs28510670) reached a suggestive significance and another five variants at *ENTHD1* gene also indicated a suggestive significance among PWH (Supplemental figure 11, <http://links.lww.com/QAD/C924>).

Different from PWH, three variants at *SCN5A* met the genome-wide significance level among HIV negative counterparts (rs7374540, $\beta = -3.23$, $P = 4.84 \times 10^{-10}$;

rs61558743, $\beta = -3.18$, $P = 6.85 \times 10^{-10}$; rs13084981, $\beta = 2.95$, $P = 1.52 \times 10^{-8}$).

QRS duration

Among PWH, 11 variants at nine genes reached genome-wide significance level to QRS duration, among which rs2307311 located at the exonic region. Five were at chromosome 3 (rs2307311, $\beta = 10.79$, $P = 8.78 \times 10^{-9}$, *MCM2*; rs76267866, $\beta = 10.97$, $P = 1.26 \times 10^{-8}$, *SAMMSON*; rs12485697, $\beta = 10.91$, $P = 1.40 \times 10^{-8}$, *SAMMSON*; rs717669, $\beta = 10.63$, $P = 2.87 \times 10^{-8}$, *SAMMSON*; rs1024889, $\beta = 10.68$, $P = 2.41 \times 10^{-8}$,

Table 2. Associations of variants with electrocardiogram parameters among people with HIV and HIV-negative controls.

CHR	SNP	POS	GENE	Location	A1	A2	N	MAF	BETA	SE	P ^a
Association with PR interval among PWH											
6	rs1144157	91184183	MAP3K7, AL391559.1	intergenic	A	G	1602	0.052	16.55	2.66	5.05E-10
8	rs115038157	144505786	MAFA, ZC3H3	intergenic	A	G	1601	0.048	16.17	2.75	4.11E-09
3	rs17043654	1807038	RPL23AP39, RPL21P17	intergenic	T	C	1601	0.049	15.76	2.71	6.12E-09
9	rs73643283	28027834	LINGO2	intronic	A	G	1602	0.064	14.02	2.44	8.89E-09
2	rs76345397	38521761	ATL2	3'UTR	C	T	1599	0.058	14.15	2.58	4.06E-08
22	rs9941936	40209077	ENTHD1	intronic	C	T	1594	0.054	14.29	2.61	4.60E-08
22	rs78556357	40140444	ENTHD1	intronic	C	A	1602	0.055	14.13	2.60	5.56E-08
22	rs73888146	40131030	ENTHD1	intergenic	T	C	1602	0.054	14.17	2.61	5.88E-08
2	rs60941472	38603021	ATL2	intronic	T	C	1602	0.058	13.91	2.57	5.94E-08
22	rs74661448	40259861	ENTHD1	intronic	T	C	1602	0.055	14.09	2.60	6.05E-08
2	rs17022549	38521908	ATL2	3'UTR	C	T	1601	0.058	13.90	2.57	6.10E-08
22	rs6001686	40206811	ENTHD1	intronic	A	C	1602	0.058	13.69	2.53	6.44E-08
22	rs73888176	40210057	ENTHD1	intronic	C	T	1601	0.054	14.05	2.61	7.19E-08
2	rs28510670	38587880	ATL2	intronic	C	T	1597	0.059	13.61	2.54	8.79E-08
Association with PR interval among HIV-negative controls											
3	rs7374540	38634142	SCN5A	intronic	A	C	3687	0.481	-3.23	0.52	4.84E-10
3	rs61558743	38631855	SCN5A	intronic	A	G	3690	0.492	-3.18	0.52	6.85E-10
3	rs13084981	38645999	SCN5A	intronic	T	C	3624	0.486	2.95	0.52	1.52E-08
Association with QRS duration among PWH											
14	rs10483994	88715733	KCNK10	intronic	G	T	1610	0.050	11.99	1.99	1.57E-09
12	rs111350614	132837066	GALNT9	intronic	C	A	1607	0.051	11.64	1.94	1.91E-09
10	rs1214549	90591896	ANKRD22	intronic	C	T	1610	0.059	10.76	1.83	3.96E-09
3	rs2307311	127336910	MCM2	exonic	A	G	1610	0.056	10.79	1.88	8.78E-09
7	rs78075968	43702505	COA1	intronic	G	A	1609	0.057	10.66	1.87	1.23E-08
3	rs76267866	70540347	SAMMSON, FOXP1	intergenic	T	A	1602	0.055	10.97	1.93	1.26E-08
8	rs75838142	69168971	PREX2	intergenic	A	G	1609	0.079	9.04	1.59	1.30E-08
3	rs12485697	70543116	SAMMSON, FOXP1	intergenic	T	C	1610	0.055	10.91	1.92	1.40E-08
3	rs1024889	70463330	COX6CP6, RP11-23113.2	intergenic	G	A	1610	0.056	10.68	1.91	2.41E-08
3	rs717669	70508727	SAMMSON, FOXP1	intergenic	G	A	1609	0.056	10.63	1.91	2.87E-08
10	rs2478830	30366521	JCAD	intronic	A	C	1610	0.066	9.74	1.75	2.88E-08
Association with QRS duration among HIV-negative controls											
16	rs11557300	420179	MRPL28	exonic	A	G	3667	0.031	4.14	0.77	6.53E-08
4	rs28558828	130222138	C4orf33, LINC02466	intergenic	T	C	3703	0.149	2.02	0.39	1.78E-07
Association with QTc interval among PWH											
3	rs9815364	149604799	RNF13	intronic	C	T	1590	0.084	-11.75	2.13	3.29E-08
3	rs2100633	149668016	RNF13	intronic	G	A	1589	0.075	-12.10	2.23	6.13E-08
3	rs1403028	149495855	ANKUB1	intronic	A	G	1590	0.095	-9.62	2.01	1.80E-06
3	rs9854382	149553226	RNF13	intronic	T	C	1539	0.077	-9.69	2.35	3.66E-05
Association with QTc interval among HIV-negative controls											
16	rs11866002	58587737	CNOT1	exonic	C	T	3612	0.428	-3.11	0.59	1.19E-07
16	rs67861272	58630500	CNOT1	intronic	G	A	3613	0.433	-3.04	0.59	2.05E-07
16	rs37060	58566304	CNOT1	intronic	A	G	3613	0.371	-3.10	0.60	2.86E-07

A1, minor allele; A2, major allele; CHR, chromosome; MAF, minor allele frequency; N, number of participants; POS, position; SE, standard error.
^aAdjusted for age, sex, tobacco use status, alcohol use status, BMI, cIMT, diabetes, hypertension, and the first five PCs.

COX6CP6) and two were at chromosome 10 (rs1214549, $\beta = 10.76$, $P = 3.96 \times 10^{-9}$, *ANKRD22*; rs2478830, $\beta = 9.74$, $P = 2.88 \times 10^{-8}$, *JCAD*). Another four variants were at genes *KCNK10*, *GALNT9*, *PREX2*, and *COA1*, respectively (rs10483994, $\beta = 11.99$, $P = 1.57 \times 10^{-9}$; rs111350614, $\beta = 11.64$, $P = 1.91 \times 10^{-9}$; rs75838142, $\beta = 9.04$, $P = 1.30 \times 10^{-8}$; rs78075968, $\beta = 10.66$, $P = 1.23 \times 10^{-8}$). (Supplemental figure 12, <http://links.lww.com/QAD/C924>)

Among HIV-negative counterparts, no variant met the genome-wide significance level.

QTc interval

Among PWH, one genome-wide significant variant was identified to be associated with QTc interval (rs9815364, $P = 3.29 \times 10^{-8}$, *RNF13*), which was in tight LD with another three suggestively significant variants at *RNF13* or

ANKUB1 (rs2100633, rs1403028, rs9854382) (Supplemental figure 10, <http://links.lww.com/QAD/C924>).

Among HIV-negative counterparts, no variant met the genome-wide significance level. Three variants at *CNOT1* gene were suggestively associated with QTc interval (rs11866002, $\beta = -3.11$, $P = 1.19 \times 10^{-7}$; rs67861272, $\beta = -3.04$, $P = 2.05 \times 10^{-7}$; rs37060, $\beta = -3.10$, $P = 2.86 \times 10^{-7}$).

The variants significantly associated with ECG parameters among PWH also had a significant effect among overall population with P value ranging between 3.19×10^{-9} and 3.60×10^{-3} (Supplemental Table 2, <http://links.lww.com/QAD/C924>).

We also measured the association of genetic variants with dichotomous ECG parameters. Details can be seen in

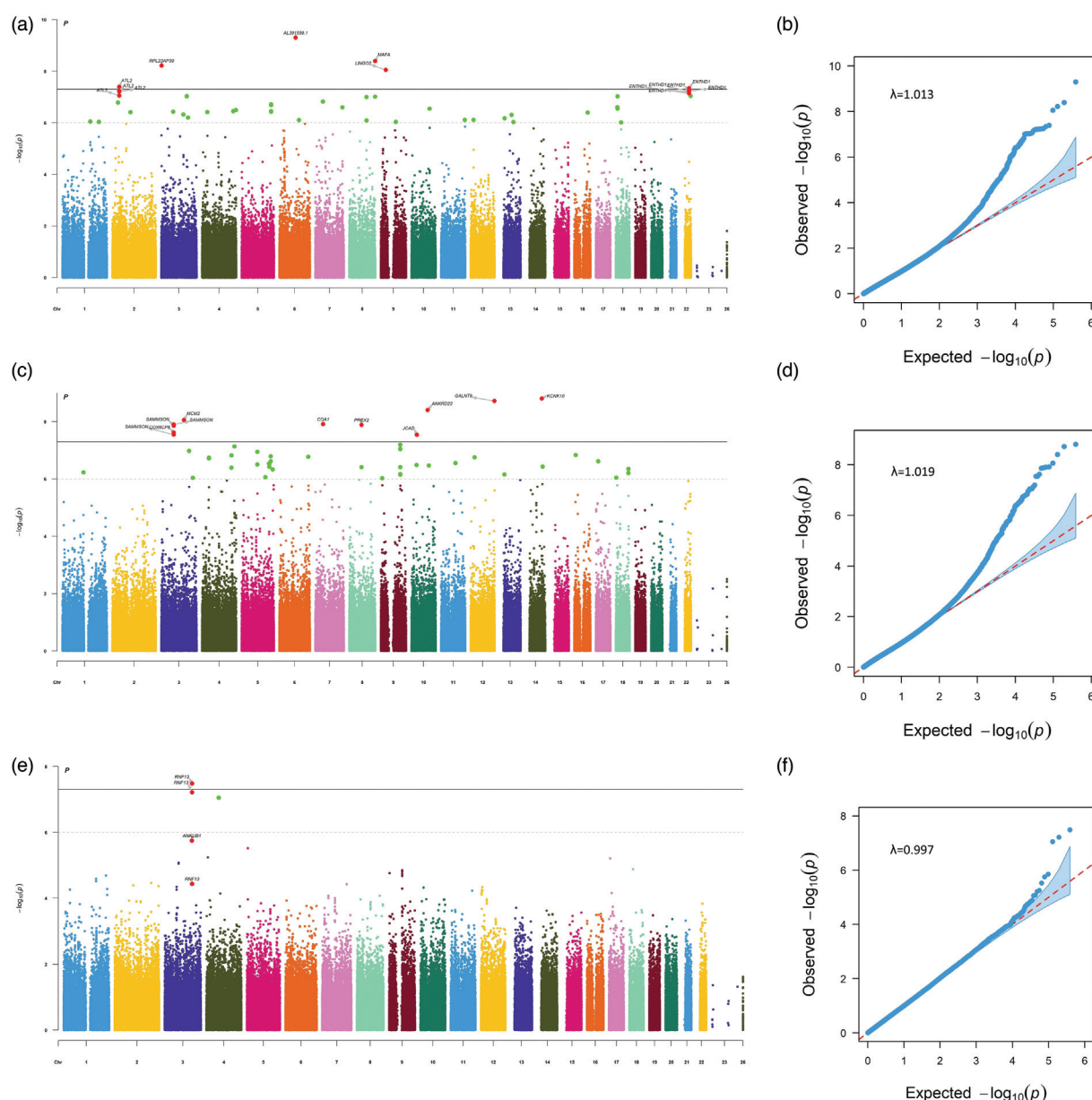


Fig. 1. Manhattan plots and qq plots of SNPs associated with PR interval (a, b), QRS duration (c, d), and QTc interval (e, f) among people with HIV. Manhattan plots showing the $-\log_{10}$ transformed two-tailed P value of each SNP from the GWAS on the y axis and base-pair positions along the chromosomes on the x axis. The red dots indicate SNPs surpassing Bonferroni-corrected suggestive significance threshold ($P < 1 \times 10^{-6}$) and SNPs in tight linkage disequilibrium with the suggestive significant SNPs.

Supplemental materials, <http://links.lww.com/QAD/C924> and Supplemental Table 3-4, <http://links.lww.com/QAD/C924>.

HIV-gene interaction on ECG parameters

For PR interval, 13 variants had a genome-wide significant interaction with HIV infection ($\beta_{\text{interact}} > 11.00$, $P_{\text{interact}} < 5 \times 10^{-8}$), among which two variants reached genome-wide significance for PR interval (rs1144157, $\beta_{\text{interact}} = 17.41$, $P_{\text{interact}} = 2.13 \times 10^{-11}$; rs9941936, $\beta_{\text{interact}} = 16.26$, $P_{\text{interact}} = 3.09 \times 10^{-10}$) and five variants at gene *ENTHD1* reached suggestive significance for PR interval among PWH in GWA analyses (rs73888146,

$\beta_{\text{interact}} = 16.14$, $P_{\text{interact}} = 3.91 \times 10^{-10}$; rs73888176, $\beta_{\text{interact}} = 16.06$, $P_{\text{interact}} = 4.42 \times 10^{-10}$; rs78556357, $\beta_{\text{interact}} = 16.05$, $P_{\text{interact}} = 4.61 \times 10^{-10}$; rs6001686, $\beta_{\text{interact}} = 15.49$, $P_{\text{interact}} = 5.31 \times 10^{-10}$; rs74661448, $\beta_{\text{interact}} = 15.75$, $P_{\text{interact}} = 9.62 \times 10^{-10}$). Both genetic variants and HIV infection were still significantly associated with PR interval in interaction analyses (Supplemental table 5, <http://links.lww.com/QAD/C924>).

For QRS duration, a total of 128 genetic variants had a genome-wide significant interaction with HIV infection, 85 located at the intergenic region, 41 at the intronic region, and 2 at the exonic region. Genome-wide

significant variants to QRS duration among PWH also had a genome-wide significant interaction with HIV infection (All $P_{\text{interact}} < 5 \times 10^{-8}$ with β ranging between 7.87 and 10.89). Both genetic variants and HIV infection were still significantly associated with QRS duration in interaction analyses (Supplemental table 5, <http://links.lww.com/QAD/C924>).

Functional annotation, gene mapping, and gene set analysis

Among PWH, we identified six genomic risk loci and 87 mapped genes associated with PR interval, eight genomic

risk loci and 79 mapped genes associated with QRS duration, and one genomic risk loci and 17 mapped genes associated with QTc interval. Among PWH, *EIF2AK2* on chromosome 2 (Fig. 2a), *PARP10* on chromosome 8 (Fig. 2b) and *APOBEC3A*, *APOBEC3C*, *APOBEC3D* on chromosome 22 (Fig. 2c) were mapped by chromatin interactions for PR interval. *ACTA2*, *LIPK*, *IFIT1*, and *IFIT5* on chromosome 10 (Fig. 2d) were mapped by chromatin interactions for QRS duration. Among HIV-negative control, one genomic risk loci and eight mapped genes were identified to be associated with PR interval. *SCN5A* gene was implicated by all three methods.

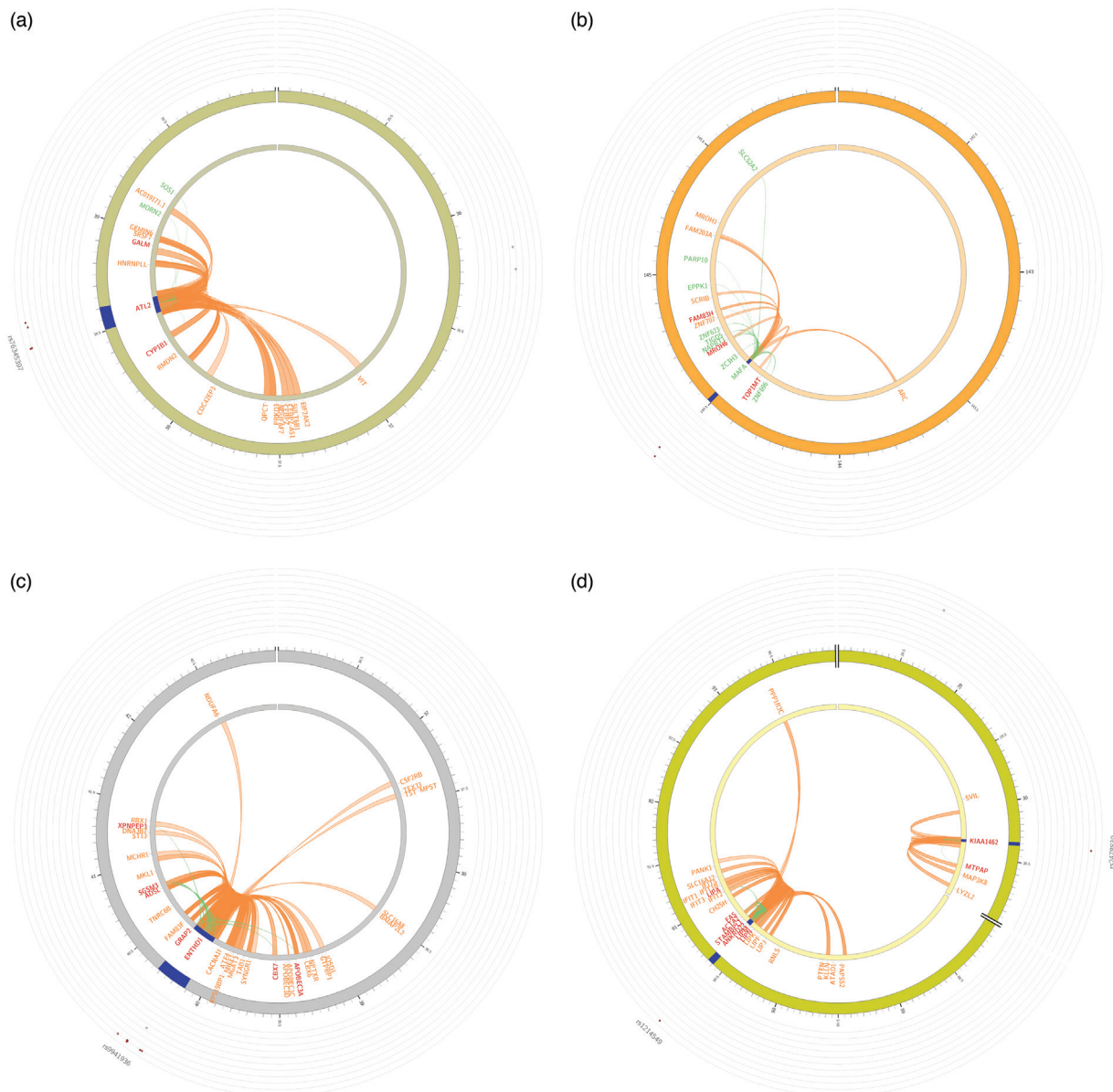


Fig. 2. Cross-locus interactions for genomic regions associated with PR interval (a, b, c) and QRS duration (d) among people with HIV. Circos plots showing genes on chromosome 2 (a), 8 (b), 22 (c), and 10 (d) that were implicated through the genomic risk loci (blue areas) by positional mapping, by chromatin interaction mapping (orange font), eQTL mapping (green font), or by both chromatin interaction and eQTL mapping (red font). The outer layer shows a Manhattan plot containing the $-\log_{10} P$ value of each SNP in the GWAS analysis of PR interval or QRS duration among PWH.

ITGA9 was mapped through eQTL mapping. The results can be seen in Supplemental table 6–8, <http://links.lww.com/QAD/C924>.

Among PWH, two most significant gene sets for PR interval were involved in the regulation of viral genome replication and DNA cytosine deamination, while the most significant gene set for QRS duration was involved in the response to virus. Among HIV-negative controls, the most significant gene set for PR interval was involved in the voltage gated sodium channel complex (Fig. 3, Supplemental table 9, <http://links.lww.com/QAD/C924>).

Discussion

Using a large, epidemiological study cohort of more than 5300 individuals of Chinese PWH and HIV-negative controls with well characterized ECG parameters, we have discovered significant associations of several

common genetic variants with PR interval, QRS duration, and QTc interval. A total of 18 genome-wide significant variants were detected among PWH, six for PR interval, 11 for QRS duration, and one for QTc interval. We have also identified variants at previously reported ECG-abnormality-related genes among general population, including *SCN5A* and *CNOT1* [26,27]. Mapped genes for PR interval among PWH were enriched in the biological process of viral genome replication and DNA cytosine deamination, while enriched pathways for QRS duration among PWH were in the biological process of response to virus and lipid catabolic process.

Among PWH, variants at *ATL2* and *ENTHD1* were associated with PR interval. The disruption of *ATLs* function leads to the formation of long, unbranched ER tubules [28] and has been linked to HIV-associated neurodegenerative diseases [29]. It has been reported that *ATL2* is required in the formation of the viral replication organelles in dengue virus and Zika virus [30]. *ATLs* are also linked to cellular processes such as lipid droplet

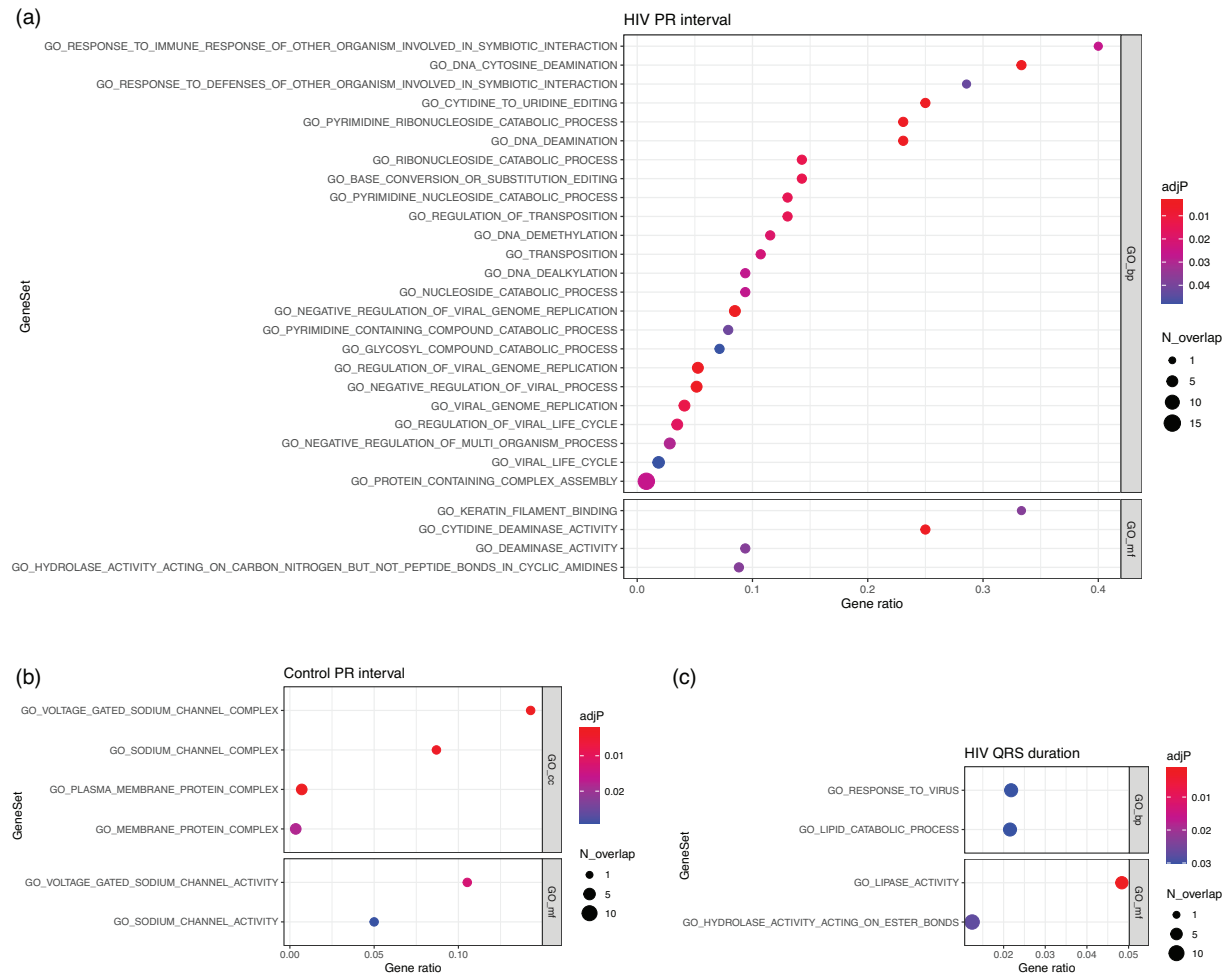


Fig. 3. Significant bubble diagrams of GO biological process, molecular function, and cell components for PR interval among PWH (a), PR interval among HIV-negative controls (b), and QRS duration among PWH (c).

dynamics, endosomal transport, selective autophagy, and inner nuclear membrane protein insertion [31–33]. HIV virus can change the electrophysiological properties of the heart and can cause inflammatory reaction in the coronary vessels [4]. We thus speculated that *ATL2* might impair cardiac function by participating in the production of HIV virus particles and in the vesicle trafficking pathway required for virion maturation. The functional feature of *ENTHD1* is largely unknown and might have a link to histone acetyltransferase activity, which was reported to participate in regulation of HIV-1 integration [34].

Variants at *KCNK10* and *JCAD* were associated with QRS duration among PWH. It has been reported that *KCNK10* mRNAs encoding stretch-sensitive $K_{2p10.1}$ channel are expressed in human heart and increased atrial *KCNK10* in heart failure patients could promote electrical reentry and atrial fibrillation, a common consequence of ECG abnormalities [35]. Previous GWAS [36] has identified an association between coronary artery disease and *JCAD* by promoting atherosclerotic plaque formation via the endothelial cell shear stress mechanotransduction pathway [37]. Atherosclerosis is a chronic inflammatory disease and some markers of inflammation were identified as predictors of recurrence of arrhythmia [38]. The close relationship between atherosclerosis and arrhythmia indicated that *JCAD* might also play a role in QRS duration variation.

For QTc interval, one significant variant rs9815364 at *RNF13* was found among PWH. *RNF13* played an important role in mediating ER stress-induced apoptosis by activating the IRE1 α -JNK signaling pathway [39]. Abnormality in ER stress response has been implicated as a contributing factor in a variety of diseases such as neurodegenerative diseases, cancer, diabetes, obesity, and viral infection [40–42]. Similar to *ATL2*, *RNF13* might influence the cardiac electrical activity by interfering with HIV viral infection and production. The major phenotypes explored in HIV GWAS were susceptibility to infection, set point viral load, and disease progression [43] and most of the reported AIDS restriction genes fall within the HLA class I locus and *CCR5* gene [44]. However, no variant identified in our analysis located in those genes, probably owing to the different ethnic populations and outcome traits.

Consistent to previous research, three variants at *SCN5A* were significantly associated with PR interval among HIV-negative controls [26]. *SCN5A* is known to affect myocardial conduction, and mutations in this gene have been associated with a number of clinical arrhythmia syndromes [26]. *SCN5A* encodes the cardiac sodium-channel $Na_v1.5$ that underlies the depolarizing current I_{Na} , and opening of the $Na_v1.5$ channel drives rapid membrane depolarization during cardiac action potential [13]. Loss-of-function of *SCN5A* reduces the excitability

of myocardial cells and slows impulse conduction throughout the myocardium [45].

Similarly, in line with previous GWAS [27], three variants at *CNOT1* reached suggestive significance for QTc interval among HIV-negative controls. *CNOT1* encodes the central subunit of CCR4-NOT, a multifunctional, conserved complex regulating gene expression and mRNA stability and turnover. Previous experimental study demonstrated that the *CNOT1* haplotype associating with longer QT-interval caused reduced reporter expression in a cardiac cell line, suggesting the reduced *CNOT1* expression may contribute to abnormal QT-intervals [46].

HIV infection had a significant interaction with genetic variants for PR interval and QRS duration, implying a possibly extensive influence of HIV infection on genetic variants association with ECG parameters. Notably, all genome-wide significant variants of QRS duration and seven genome-wide or suggestively significant variants of PR interval among PWH had a significant interaction with HIV infection. This also indicated that HIV infection might play a crucial role in the cardiac electrical activity apart from the host genome and jointly determine the disease outcomes. The underlying mechanism might be attributable to the integration of HIV proviral DNA into host genome, which could affect expression of host genes, influence basal and inducible transcription, and thus manifest differential association of genetic variants with ECG parameters for PWH and HIV-negative controls.

HIV virus in particular favors the interior regions of genes that reside in relatively gene-dense or transcriptionally active regions of chromatin when choosing integration site [47], which might cause alteration of host gene expression in PWH and different genetic variants associated with ECG parameters from negative controls. Using Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis [48], researchers have demonstrated that HIV integration sites in clonally expanded cells were in pathways associated with evasion of apoptosis and cell proliferation [49]. As a response, host genome might defend against the HIV integration by altering gene expression, which is another possible reason of different genetic variants associated with ECG parameters between PWH and negative controls. Nonetheless, the exact mechanisms require further investigation.

For PR interval among PWH, most genes were enriched in the regulation of viral genome replication, DNA deamination and regulation of viral life cycle. Host genome might restrict HIV viral replication by altering gene expression in those pathways, finally influencing cardiac electrical activities. One mostly involved gene, *APOBEC3A*, mapped by both eQTL and chromatin interaction mapping, plays a role in immunity by restricting transmission of foreign DNA such as viruses

[50]. *APOBEC3A* could also maintain and reinforce HIV-1 latency [51]. Similarly, genes related to QRS duration among PWH were enriched in the response to virus and lipid catabolic process, the latter of which might participate in the production of HIV viral membrane. This further suggested that host genome might affect the ECG parameters through influencing HIV viral activities.

Without the influence of HIV replication and integration, mapped genes among HIV-negative controls for PR interval were enriched in the cellular component of voltage gated sodium channel complex and plasma membrane protein complex. Voltage-gated sodium channels are key regulators of cellular excitability and strongly involved in the generation of action potentials, heart contraction, and in the sensing of pain [52]. In cardiomyocytes, the rapid depolarization of the membrane potential is mediated by the cardiac voltage-gated sodium channel, encoded by *SCN5A* which was involved in all enriched pathway [52]. *ITGA9* was involved in the cellular component of plasma membrane protein complex and implicated in atrioventricular conduction and might mark a distal *SCN5A* regulatory element [53].

To our knowledge, this is the first and largest GWAS of measuring genetic variants associated with ECG parameters among Chinese PWH and HIV-negative controls. Nevertheless, several limitations warrant further consideration. First, replication study was not conducted given difficulties in identifying additional studies of Chinese PWH with ECG measures and genotype data, which may reduce the robustness of our results to some extent. However, using the stringent *P* value could reduce the false discovery rate and candidate SNPs and genes were presented for future validation. Second, no imputation of SNP genotype was performed, as all genetic data were available within one cohort and were obtained using a single chip. However, all the association results were based on the actual measurement of genetic variants, which may make sure the authenticity of GWA analyses. Lastly, the impact of higher ECG abnormality prevalence among HIV-negative controls on genetic association analyses also warrants further exploration.

Conclusion

The present GWAS indicated a distinctive impact of host genome on quantitative ECG parameters among PWH, as well as significant interaction effects between HIV infection and genetic variants on ECG parameters. Different from HIV-negative controls, host genome might influence the cardiac electrical activity by interfering HIV viral infection, production, and latency among PWH. Better understanding of the biologic pathways and molecular genetics associated with cardiac conduction may offer insight into the pathogenesis of conduction abnormalities among PWH.

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N.H. proposed the research question and generally supervised the study. J.H. contributed to laboratory work, data analysis, and manuscript drafting. H.L. and Y.D. supervised individual enrollment, sample collection, and data management. X.L. contributed to genomic data collection, management, and analysis. X.C., W.S., S.Z., M.W., and J.X. contributed to data collection and performing experiments. J.H., Y.D., X.L., H.L., and N.H. had full access to all the data. All authors critically reviewed and edited the manuscript and consented to final publication.

No competing interests are declared.

The sponsors of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

All the data in the article or in the supplementary materials are free to obtain. Raw data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of interest

There are no conflicts of interest.

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